

AURA PLATFORM CONTROL & API MANUAL

Comprehensive Super Admin Operations, Tenant Isolation, & Developer Integrations

Platform Owner Metadata

- 🔑 Role Profile: Platform Super Administrator / System Owner
- 📧 Target Workspace Address: lztalaslamqadm@gmail.com (Root Principal)
- 🔑 Access Level: Comprehensive Database, API, Backups & Subscriptions Control
- 🏠 Tenant Model: Logical Multi-Tenant database virtualization via Firestore
- 🔒 Security Level: Zero cross-tenant data leak guarantees via dynamic API keys
- 🔗 Integration Protocols: REST, JWT Authorization Bearer, Custom Headers

MANUAL INDEX

- Section 1: Multi-Tenant Architecture & Database Isolation Model
- Section 2: API Gateway Management & University Key Issuance
- Section 3: Yemen Mobile Wallet Subscriptions & Financial Management
- Section 4: Enterprise Backups, Point-In-Time Restore & Logs
- Section 5: Complete Operational Step-by-Step Scenario

1. MULTI-TENANT ARCHITECTURE & DATABASE ISOLATION MODEL

AURA operates on a highly isolated logical multi-tenant database pattern. Instead of physical database segmentation which incurs significant infrastructure cost, AURA virtualizes databases using institutional identifiers (e.g. "oxford-global", "cairo-national"). This prevents structural interference and ensures secure isolation:

🔗 ½ Logical Partitioning: Every database transaction, record update, and analytical query requires an `institutionId` parameter. The system strictly filters local state queries with this key.

🔗 ½ UI Workspace Isolation: When you log in as Platform Admin, you can switch between institutions via the active workspace picker in the side panel. All lists, dashboards, and reports instantly filter to the selected institution, protecting data hygiene.

🔗 ½ Public Port and Route Restrictions: The server binds specifically to port 3000 on 0.0.0.0. Student evaluations run via short code parameters, immediately resolving the correct institution record from memory.

2. API GATEWAY MANAGEMENT & UNIVERSITY KEY ISSUANCE

As Platform Admin, you control the REST API gateway through which partner universities connect their Learning Management Systems (LMS - e.g., Moodle, Blackboard) or Student Information Systems (SIS).

How to Issue API Credentials to a University:

1. Go to the Developer API Console in the sidebar.
2. The system automatically provisions credentials specifically for the active institution.
3. Copy the following issued tokens and hand them over to the university's IT Department:
 - `x-api-key`: Unique Client Key used in request headers to authorize the institution.
 - `Client Secret`: Cryptographic key to verify payload authenticity.
 - `Authorization JWT`: Pre-signed Bearer Token with encrypted institution parameters.
4. The university developer must include these headers in their curl/fetch requests:
Headers: { `"x-api-key": "...", "Authorization": "Bearer ...", "x-institution-id": "..."`

Malicious Cross-Tenant Attack Safeguards:

To simulate security attacks, press "Simulate Attack" in the API playground. This tests for cross-tenant data leakage (e.g., using Oxford's API key while requesting Cairo's private records). AURA authenticates the request and returns an immediate 401 Unauthorized block, guaranteeing full security.

3. YEMEN MOBILE WALLET SUBSCRIPTIONS & FINANCIALS

AURA features a highly targeted regional mobile wallet subscription billing system designed for Yemen. This coordinates subscription flows via leading mobile networks including "Jawali" and "Jeeb".

A. How to Configure Platform Wallets:

- 1. Navigate to the Super Admin Dashboard (accessible only to Platform Admin users).
- 2. Scroll down to the "Regional Wallet Configuration" panel.
- 3. Enter your active merchant wallet details: Wallet Phone Number, Account Name, and Merchant API ID.
- 4. Press "Update Configuration" to instantly apply the new credentials to the platform checkout.

B. Understanding the Payment Review & Approval Pipeline:

- ١٤١٢ Student campaigns, premium AI reports, and maximum limits are governed by SaaS Subscriptions.
- ١٤١٢ When a tenant university requests a subscription upgrade (e.g. Basic to Enterprise), they select a plan and receive your configured Wallet details. They transfer the payment and submit proof of transaction (Reference Number and optional receipt snapshot).
- ١٤١٢ The payment record is captured in the "Pending Subscriptions Audit" queue on your Super Admin panel.
- ١٤١٢ You can review the payment detail, verify the transfer on your merchant phone, and press:
 - Approve: Automatically updates the university subscription to "Active", sets limits, and unlocks API/AI.
 - Reject: Suspends active upgrades and requests the tenant to submit a valid transaction token.

SAAS PLAN PRICING & QUOTA MATRIX

Plan Name	Yemeni Rial Equivalent	Max Users	Max Students	Max Lecturers	API / AI Access
Basic	300,000 YER / Month	30	5,000	300	Disabled
Professional	600,000 YER / Month	100	15,000	1,000	Enabled
Enterprise	1,200,000 YER / Month	200	30,000	2,500	Full Features

4. ENTERPRISE BACKUPS, POINT-IN-TIME RESTORE & AUDIT LOGS

AURA ensures unmatched enterprise durability by providing double-layer backup and restoration tools.

These are accessible via the Super Admin Dashboard and operate in real-time without database downtime.

A. Point-In-Time Recovery (PITR) Backup System:

🔧 **On-Demand Snapshot:** Click "Create Full System Backup" in the Admin controls. The server takes a complete JSON state mirror of all tenant records, databases, evaluation parameters, and wallet settings.

🔧 **Timestamp Tagging:** Backups are saved with precise ISO-timestamps and labeled with metadata (e.g. "Pre-Upgrade") to maintain clean recovery logs.

🔧 **Instant Point-in-Time Restore:** If a database error or configuration leak occurs, scroll to the backups list, locate your target snapshot, and press "Restore". The system instantly replaces active memory structures and updates persistent layers seamlessly.

B. Live Audit Logging & IP Security Tracing:

AURA maintains a strict compliance audit ledger capturing two distinct log types:

1. **Operational Audit Logs (Tenant Specific):** Track academic actions (e.g., submitting self-study, creating courses, closing evaluations) accessible by local institution quality managers.
2. **Live Server Security Logs (Platform Wide):** Accessible only on the Super Admin panel. These record low-level server events, tracking: timestamp, client IP address, action performed, tenant institution, user identity, and detailed execution parameters. This satisfies regional accreditation requirements for data security auditing.

5. STEP-BY-STEP PLATFORM USER GUIDE & OPERATIONAL RUNBOOKS

As Platform Admin, execute the following runbooks for daily administrative procedures:

Runbook A: Onboarding a New University Tenant

1. Open the Super Admin Dashboard and locate the "Institutions Directory" panel.
2. Click "Add New Institution". Fill out parameters: English/Arabic Name, active domain name, and subscription tier.
3. Under the User Manager section, create a new "Quality Manager" or "Dean" account assigned specifically to the new institution's tenant ID.
4. Log out and instruct the university's Dean to log in. Their workspace starts completely empty, fully isolated and secure.

Runbook B: Managing and Approving Campaign Evaluations

1. In the sidebar, select "Evaluation Forms & Campaigns".
2. You can review drafts of campaigns prepared by local college coordinators.
3. Transition the campaign status: "Draft" -> "Under Review" -> "Approved" -> "Active".
4. Once marked "Active", the student evaluation gateway is unlocked. Share the 8-digit access code (e.g. EVAL-793X) with students to collect anonymous, aggregated responses.

Runbook C: Executing AI Audits & Compliance Reviews

1. Navigate to the "AI Compliance Advisor" or "Quality Evaluations".
2. Click "Generate Strategic Quality Audit" or "AI Recommendation Review".
3. The server securely proxies isolated university metrics to the Google Gemini model using the secure GEMINI_API_KEY environment secret, returning detailed compliance advice and actionable steps.

SYSTEM OWNER AUTHORIZATION NOTICE

This operations manual is cryptographically bound to the principal admin account. Unsanctioned duplication or sharing of this document is strictly prohibited under SaaS compliance parameters.